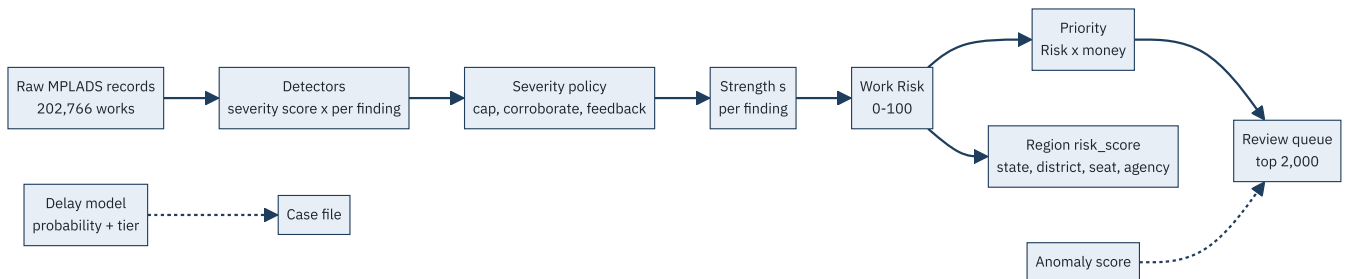


MPLADS Review — Risk Scoring Formulas

Aaryan Jain · 24 September 2026 · Data snapshot of 10 September 2026

Every work gets two independent signals: a Risk score (0–100) from rule-based detectors, and a delay probability from a machine-learning model. Only the rules create flags; the models rank and advise.

How the pieces fit



Solid arrows are the rule-based engine, which alone creates flags. Dotted arrows are the models: they advise on a work's case file and reorder the queue, but never add, remove or regrade a flag.

Rule-based engine, step 1: severity score of each finding

Each finding gets a severity score x between 0 and 1: how far past its threshold the work is. The label is a band of x , and the detector's own rule still decides the label; x only places the finding inside its band.

low : $0 \leq x < 0.3333$ medium : $0.3333 \leq x < 0.6667$ high : $0.6667 \leq x \leq 1$

x rises linearly between the points listed (a piecewise-linear ramp):

Detector	What is measured	x rises from → to
Peer-comparison delays (Delay in Sanction, Recommendation Pending Sanction, Sanctioned Work Not Taken Up, Delay in Completion, Work Not Completed, Payment Without Completion)	days, against peers in the same state × activity × term	peer P90 → 0, P95 → $\frac{1}{3}$, P99 → $\frac{2}{3}$, $(2 \cdot P99 - P90) \rightarrow 1$
Prolonged Delay Beyond Guideline (fixed limits: 365 days to sanction, 365 days unsanctioned, 540 days open)	days ÷ limit	$1.0x \rightarrow \frac{1}{3}$, $1.5x \rightarrow \frac{2}{3}$, $3x \rightarrow 1$
Excess Expenditure; Sanction Exceeds Recommendation	paid ÷ sanctioned, or sanctioned ÷ recommended	$1.01 \rightarrow \frac{2}{3}$, $2.0 \rightarrow 1$
Unusual Cost	robust z of $\log_{10}(\text{cost})$ vs state × activity × term peers	$z \ 5 \rightarrow 0$, $6.5 \rightarrow \frac{1}{3}$, $13 \rightarrow \frac{2}{3}$, $19.5 \rightarrow 1$
Expenditure After Completion	days from completion to last payment	$30 \rightarrow 0$, $180 \rightarrow \frac{1}{3}$, $540 \rightarrow \frac{2}{3}$

Detector	What is measured	x rises from → to
Concentration of Works in a Single Agency	agency's share of district payments	tenure P90 → 0, P95 → 1/3, P99 → 2/3, 100% → 1
Payment and Completion Mismatch	paid – completed ÷ completed	0.05 → 0, 0.5 → 1/3
Trust/Society Limit Exceeded	total ÷ ₹1 crore cap	1.0× → 1/3, 2× → 2/3
Allocation Limit Exceeded	overage ÷ allocation	0.05 → 1/3, 0.5 → 2/3
Unspent Balance	1 – utilisation	0.4 → 0, 1.0 → 1/3
Fixed-severity detectors (Expenditure Without Sanction, Duplicate Work, Prohibited Work, Completion Evidence Not Attached, Record Date Discrepancy, Calamity Consent)	–	centre of their band: 1/6, 1/2 or 5/6

The robust z for cost is:

$$z = \frac{0.6745 (\log_{10} \text{cost} - \text{median}_{\text{peers}})}{\max(\text{MAD}_{\text{peers}}, 0.05)}$$

When both delay checks flag the same delay on one work, they merge into one finding: the specific tag is kept, it counts as rule evidence, and its score is the higher of the two (11,561 duplicates merged).

Rule-based engine, step 2: strength, Risk and Priority

A work's Risk combines its strongest finding from each family (timing, money, guideline, concentration, documentation, data integrity), and Priority scales Risk by the money at stake.

Severity adjustments. A statistical finding above the middle of medium is compressed into the top half of medium, which keeps the order among strong outliers:

$$x' = 0.5 + (x - 0.5) \cdot \frac{0.6666 - 0.5}{1 - 0.5} \quad (x > 0.5)$$

If a work has findings in two or more families (data-entry findings excluded), its strongest statistical finding moves up one band:

$$x' = \min\left(1, x + \frac{1}{3}\right)$$

Officer reviews then apply: a tag with 50 or more reviews and a Wilson lower bound below 0.7 cannot be high, and a tag-plus-agency or tag-plus-state pattern with 10 or more reviews and at least 80% dismissed is hidden.

Strength of each finding. The curve is linear through five points, passing through the old fixed weights at each band's centre:

$$s = \text{curve}(x) \times c, \quad \text{curve} : (0, 0.10) \left(\frac{1}{6}, 0.15\right) \left(\frac{1}{2}, 0.35\right) \left(\frac{5}{6}, 0.60\right) (1, 0.70)$$

Confidence c is 1.0 for rule findings and 0.6 for statistical ones. Data-integrity findings are capped at $s = 0.05$.

Risk of each work (0–100). Only the strongest finding per family f counts, so three delay tags on one stalled work don't stack:

$$\text{Risk} = 100 \left[1 - \prod_f (1 - s_f^{\max}) \right]$$

Priority of each work. E grows from 0 at ₹1 lakh to 1 at ₹1 crore:

$$E = \min \left(1, \max \left(0, \frac{\log_{10}(\text{amount}/100,000)}{2} \right) \right)$$

$$\text{Priority} = \text{Risk} \times (0.5 + 0.5E)$$

The amount is the work's sanctioned amount, or its recommended amount if not yet sanctioned. The review queue takes 18th Lok Sabha works of ₹5 lakh or more, top 2,000 by Priority.

Worked example. A ₹70 lakh work has a rule-based high timing finding ($x = 0.83 \rightarrow s = 0.60$) and a statistical medium money finding ($x = 0.5 \rightarrow s = 0.35 \times 0.6 = 0.21$). $\text{Risk} = 100 \times [1 - 0.40 \times 0.79] = 68.4$. $E = \log_{10}(70) \div 2 = 0.92$, so $\text{Priority} = 68.4 \times 0.96 = 65.8$.

Rule-based engine, step 3: region risk scores

A state's, district's, constituency's or agency's `risk_score` is its adjusted flagged rate times the money-weighted Risk of its flagged works. It never adds up priorities, which would just rank places by size.

Only substantive works count as flagged: a work whose only findings are missing documents or data-entry mismatches is left out of these counts (it still shows on its own case file).

Adjusted flagged rate. Small regions are pulled toward the overall rate p_0 , by an amount m estimated from the data (empirical Bayes, beta-binomial method of moments):

$$p_0 = \frac{\sum_i k_i}{\sum_i n_i} \quad \tau^2 = \frac{\sum_i n_i (k_i/n_i - p_0)^2 - (R - 1) p_0 (1 - p_0)}{N - \sum_i n_i^2 / N} \quad m = \frac{p_0 (1 - p_0)}{\tau^2} - 1$$

Here k_i is flagged works and n_i is all works in region i , R is the number of regions, and $N = \sum n_i$. m is limited to 5–1,000; it comes out at 80 for states, 7.4 for districts and 8.0 for constituencies.

$$\text{shrunk_rate}_i = \frac{k_i + p_0 m}{n_i + m}$$

Money-weighted Risk and the final score:

$$\text{weighted_risk}_i = \frac{\sum_{w \in i} \text{amount}_w \cdot \text{Risk}_w}{\sum_{w \in i} \text{amount}_w}$$

$$\text{risk_score}_i = \text{shrunk_rate}_i \times \text{weighted_risk}_i$$

The breach rate shown on dashboards is the plain $k_i \div n_i$.

Delay prediction model

On the day a work is recommended, the model gives the probability that it will be seriously delayed. It uses only information known that day, and is trained on 90,838 17th Lok Sabha works recommended from 1 April 2023 (the eSAKSHI go-live; earlier records are only the unfinished works carried over).

Delay gates, the 90th percentiles of the two lags in the training works:

$$G_s = P_{90}(\text{sanction date} - \text{recommendation date}) = 382 \text{ days}$$

$$G_e = P_{90}(\text{completion date} - \text{sanction date}) = 702 \text{ days}$$

Label of each training work. t is the snapshot date (10 September 2026):

$$y = \begin{cases} 1 & \text{sanction lag} > G_s \vee (\text{unsanctioned} \wedge t - \text{rec. date} > G_s) \\ & \vee \text{execution lag} > G_e \vee (\text{open} \wedge t - \text{sanction date} > G_e) \\ 0 & \text{sanctioned and completed, sanction lag} \leq G_s, \text{ execution lag} \leq G_e \\ \text{excluded} & \text{otherwise (too young to judge; 2,062 works)} \end{cases}$$

The 12 inputs, ranked by how much the model relies on each (permutation importance, drop in AUC when the input is shuffled):

Rank	Input	Formula	Importance
1	District frequency	works in this district $\div$ all training works	0.080
2	State frequency	works in this state $\div$ all training works	0.077
3	Amount	$\log_{10}(\text{recommended amount})$	0.035
4	Activity frequency	works of this activity $\div$ all training works	0.028
5	Letter batch size	works sharing this recommendation letter number	0.011
6	Description length	characters in the work description	0.010
7	Authority workload	works the same district authority received in [date – 90 days, date)	0.009
8	Relative cost	$\log_{10}(\text{amount}) - \text{median } \log_{10}(\text{amount})$ of peers (same state and activity if ≥ 5 works, else same state, else all)	0.003
9	MP workload	works the same MP recommended in [date – 30 days, date)	0.003
10	Month	month of the recommendation date	0.003
11	Year-end window	1 if January, February or March, else 0	0.000

Rank	Input	Formula	Importance
12	Quarter	calendar quarter of the recommendation date	0.000

Model and probability. Gradient-boosted decision trees (600 trees, learning rate 0.05, up to 31 leaves per tree, at least 40 works per leaf, L2 regularisation 1.0), then isotonic calibration g fitted on scores for works each tree set never trained on:

$$p = g \left(\sigma \left(\sum_{k=1}^{600} f_k(\mathbf{x}) \right) \right)$$

Here f_k is the k -th tree and σ the logistic function. After calibration, $p = 0.64$ means about 64% of such works were delayed.

Tiers, set from the model's own history:

Tier	Cut-off on p	Share of past works	Actually delayed
High	≥ 0.641	riskiest 10%	69%
Medium	≥ 0.391	next 30%	52%
Low	< 0.391	remaining 60%	23%

Reasons shown on the case file. Going down the importance ranking, an input is listed when this work's value is at or above the 90th percentile, or at or below the 10th, of all training works (up to 3 reasons).

Accuracy, on works the model never saw:

Test	AUC	PR-AUC	Top-10% hit rate
Unseen constituencies (5-fold, grouped)	0.731	0.586	0.693
Forward in time (train before 2 Feb 2024, test after)	0.734	0.584	0.697

The base rate is 36% delayed. Recall at $p \geq 0.5$ is 0.478, and the Brier score is 0.195.

Ranking aids: anomaly score and rule-agreement score

Neither of these creates, removes or regrades a flag.

Anomaly score reorders works only inside the same 5-point Priority band of the review queue. One Isolation Forest is fitted per lifecycle stage (recommended, sanctioned and open, completed) $\times$ cost quarter, 12 groups in all, so each work is compared only with works at the same point in their life:

$$\text{anomaly}_w = \frac{\#\{v \in \text{group}(w) : \text{score}(v) > \text{score}(w)\}}{|\text{group}(w)|}$$

In words, it is the share of works in the same group that look more normal than this one, so 0.95 means more unusual than 95% of its peers.

Rule-agreement score is a gradient-boosted classifier on 19 raw inputs (amounts, ratios, lifecycle gaps, disbursement counts), trained to recognise works the rules graded high. Its AUC of 0.981 (grouped by

constituency) measures how well it reproduces the rules, not real-world risk, so it is shown only as context on the case file.

Reviewer-label model. A calibrated classifier trains on officers' Verified/Dismissed verdicts once 300 reviews exist (0 so far). It will be the only model that learns real outcomes rather than the rules.